

SYSTEMS ANALYSIS

Systems Analysis & Design

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FRANCISCO JOSÉ DE CALDAS

Outline

1 Basic Concepts /

2 Chaos and Dynamic Systems /

3 Abstraction and Modularity /



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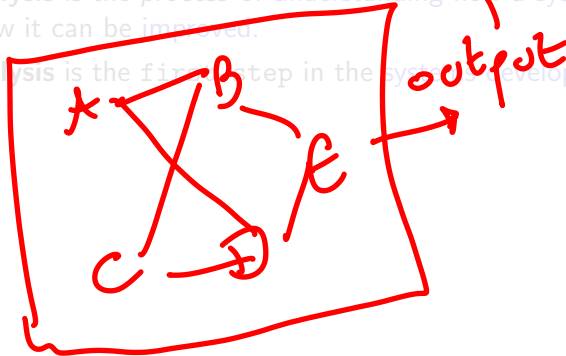
What is a System?

- A **system** is a set of interacting components that work together to achieve a common goal.
- A **system** is a collection of elements organized in a specific way.
- A **system** is a structure designed to perform a specific function.



Systems Analysis Process

- **Systems analysis** is the *process* of **studying** a **system** in order to identify its **components**, **interactions**, and **goals**.
- Systems analysis is the *process* of understanding how a system works and how it can be improved.
- Systems analysis is the first step in the system development lifecycle.



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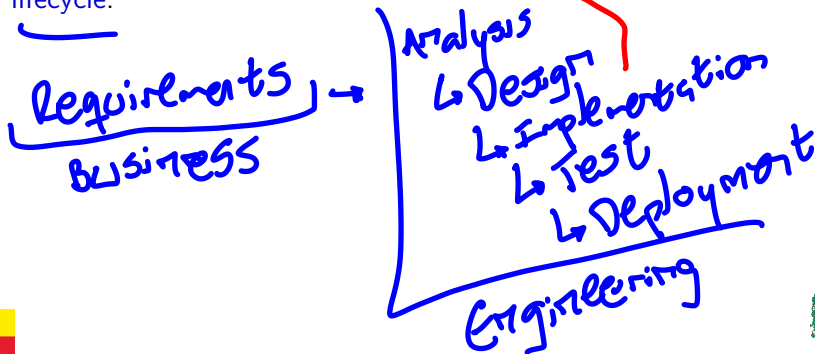
engineering

structure
responsabil: ties
sensitivity



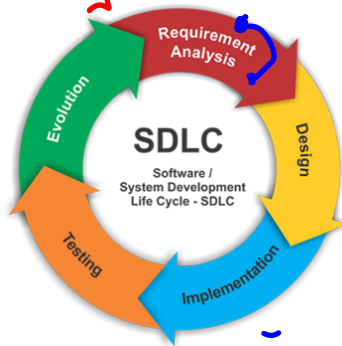
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Systems Development Lifecycle

- The **systems development lifecycle** is a *process* that *guides* the development of a *system*.
- It includes *planning*, *analysis*, *design*, *implementation*, and *maintenance* phases.
- It is a **structured approach** to developing a *system* that *ensures* it meets the *needs* of its *users*.



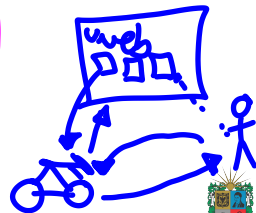
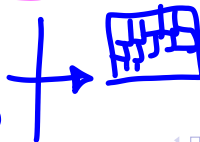
Systems Analysis Techniques

- **Systems analysis** uses a variety of techniques to study a system.
- These include interviews, surveys, observations, and document analysis.
- It also includes data modeling, process modeling, and requirements analysis.

interaction
improvisation

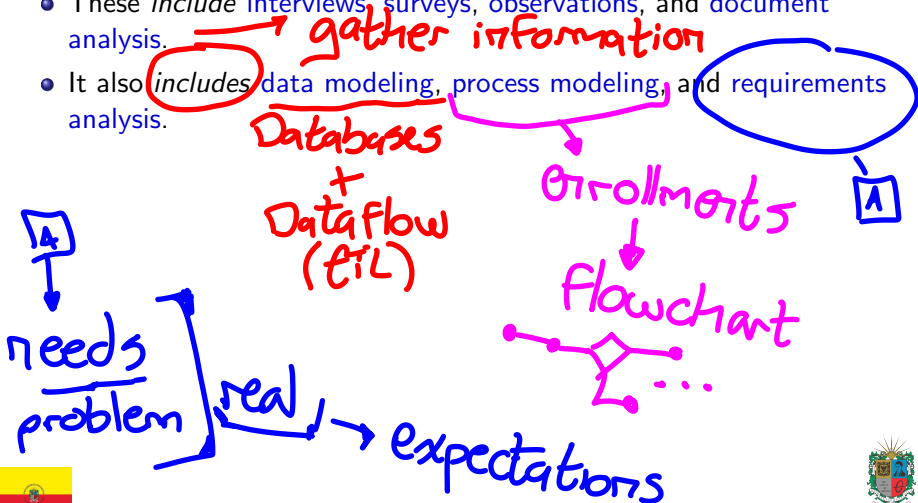
questions
↓
quick
↓
tabular
↓
analysis

Netflix
1997
Internet
↑
startup



Systems Analysis Techniques

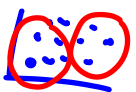
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Systems Analysis Tools

- **Systems analysis** uses a *variety* of **tools** to **study** a system.
- These *include* diagrams, charts, flowcharts, and data models.
- It also *includes* software tools such as spreadsheets, databases, and simulation software.

draw
adjust
explain
experts



Business
Intelligence

Business
Process

DB
ETL
Discovery

represent



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time → random → chaos
 → sensitivity
 → emergent behaviors



SQL
 NoSQL
 Graph
 Vector
 Doc
 $X=V$
 ...



Lateral Thinking

- **Lateral thinking** is a **creative** problem-solving technique that involves **thinking outside the box**.
- It is a **non-linear** approach to **problem-solving** that *encourages* **innovation** and **creativity**. **→ Engineers**
- It is a *useful* **technique** for *generating* **new ideas** and **solving complex problems**.
- Examples:
 - How can you improve the design of a product?
 - What are the benefits of failure?
 - Why is ignorance important?
 - When is failure better than success?

de
curve
rel.



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Coca-Cola Y-3000

Microsoft

startups → wall-e



Lateral Thinking Training

- **Lateral thinking** is a *skill* that can be learned and developed through training and practice.
- It involves exercises, games, and activities that encourage creative thinking.
- Examples of lateral thinking exercises:
 - Brainstorming sessions.
 - Mind mapping exercises.
 - Role-playing games.
 - Problem-solving activities.

① Minecraft
② Programming Contest
- 1 seg
- 1 kb



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sudoku
puzzle
tetris
go

Interviews

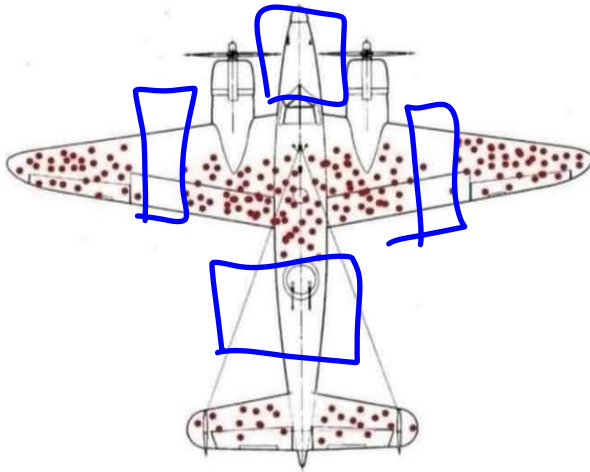
chess

serious games



Lateral Thinking Exercise

World War II, the perspective of Abraham Wald:



Uncertainty and Risk

- **Uncertainty** is the lack of knowledge about the future outcome of a decision or event.

- Risk is the probability of a negative outcome or loss associated with a decision or event.
- Uncertainty and risk are inherent in complex systems and decisions.
- They can be managed through planning, analysis, and mitigation strategies.

predictive
analysis

simulations

machine
learning
↓
probability



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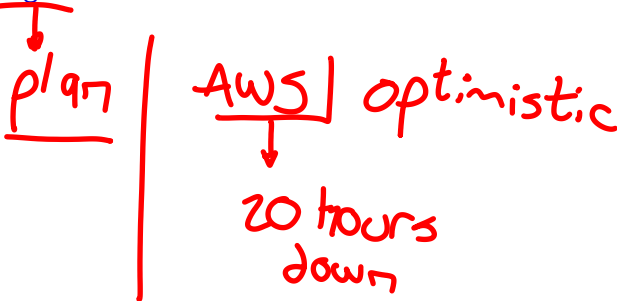
↳ A.I. → Prompt Injection

LLM
 NN → billion
 neurons $\left[\frac{ax+by}{\text{neuron}} \right]$



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Complexity and Emergence

- **Complexity** is the **degree** to which a **system** is **difficult** to **understand**.
- Emergence is the appearance of unexpected properties in a system that arise from the interactions of its components.
- Complexity and emergence are common in dynamic systems that are non-linear and chaotic.
- They can be studied and understood through systems analysis and modeling.

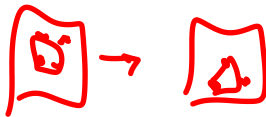


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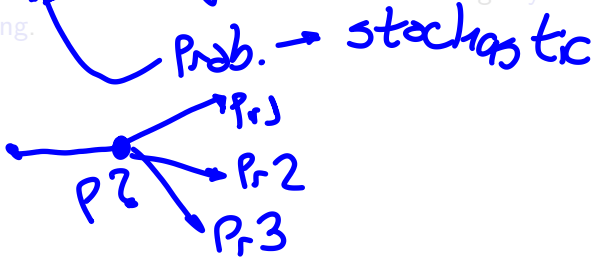
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Simulation-time



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↳ tech ⇒ programming



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What is Chaos?

- **Chaos** is a branch of mathematics that studies the sensitivity of dynamical systems to initial conditions.
- **Chaos** is a non-linear behavior that is highly sensitive to initial conditions.

→ Chaos is a deterministic behavior that is not predictable in the long term.

→ Chaos is a complex behavior that is hard to understand.



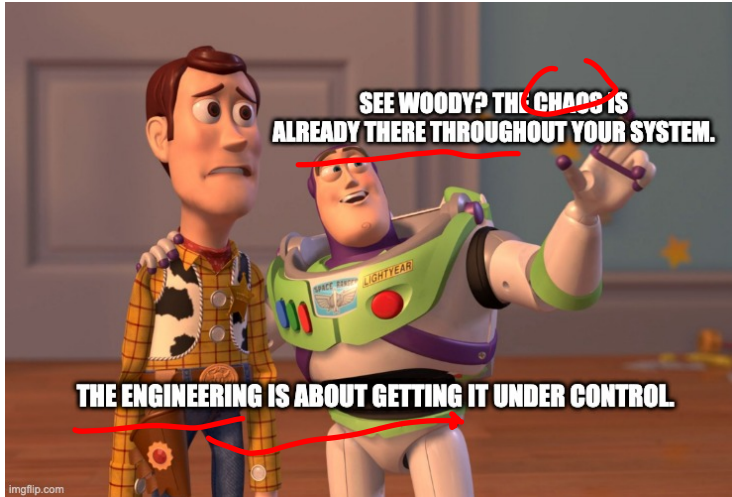
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Order ————> Disorder



Chaos is Everywhere!



What is a Dynamic System?

- A dynamic system is a system that changes over time t
- A dynamic system is sensitive to initial conditions.
- A dynamic system is non-linear.
- A dynamic system can be chaotic.



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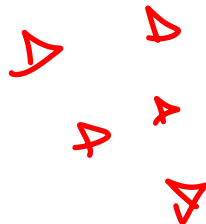
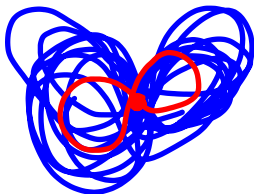


Chaotic Attractors

A **chaotic attractor** is a set of points in a phase space that attracts the trajectories of a dynamical system. One of the most famous chaotic attractors is the Lorenz attractor, discovered by *Edward Lorenz* in 1963 while studying a simplified model of atmospheric convection. It is defined by the following set of differential equations:

$$\frac{dx}{dt} = \rho(y - x)$$

$$\frac{dy}{dt} = x(\rho - z) - y$$



Fractals in Nature

2.2, 2.8, 1.7, ...

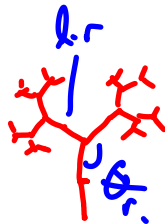
Fibonacci

1326

fractional



Lorenz system



Watch this video: <https://www.youtube.com/watch?v=kkGeOWYOFoA>



Swarm Intelligence I

ejemplo

- **Swarm Intelligence** is the collective behavior of **decentralized, self-organized systems**, natural or artificial.
 - The concept is employed in work on **artificial intelligence**.
 - The expression was introduced by Gerardo Beni and Jing Wang in 1989, in the context of cellular robotic systems. For example, watch this **video**.
- artificial life



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low
cost



Swarm Intelligence II

Useless

- The idea is: if you observe an individual, it may seem random; however, several individuals interacting with each other and the environment show intelligent behaviors.
- There are interesting population behaviors in nature, especially in insects: bees, ants, termites, among others.
- There are also many examples in nature: schools of fish, birds, wolves.

emergent



Swarm Intelligence II

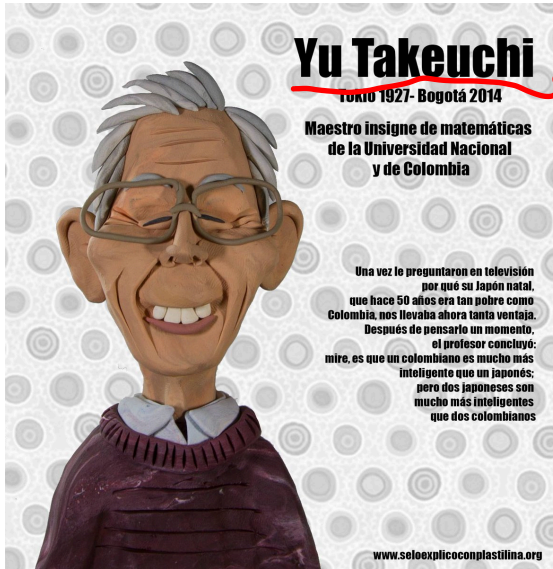
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bee
ant
fish

processing
capability



Swarm Intelligence III

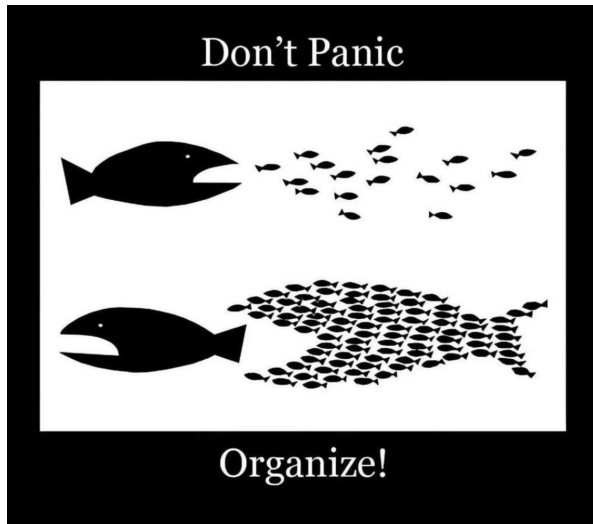


Emergent Behaviors

- Emergent behavior is the **appearance** of **complex patterns** and behaviors from **many** simple interactions.
- Emergent behavior **results** from the **collective** behavior of the **individuals** in the system.
- Emergent behavior is **not planned** or **designed** by any individual, but **arises** from their **interactions**.
- Emergent behavior is **not** the **sum** of the **individual** behaviors, but **something more**: **synergy**.
- **Swarm intelligence** refers to interesting **emergent behaviors**.



School Fish Algorithm



School Fish Algorithm

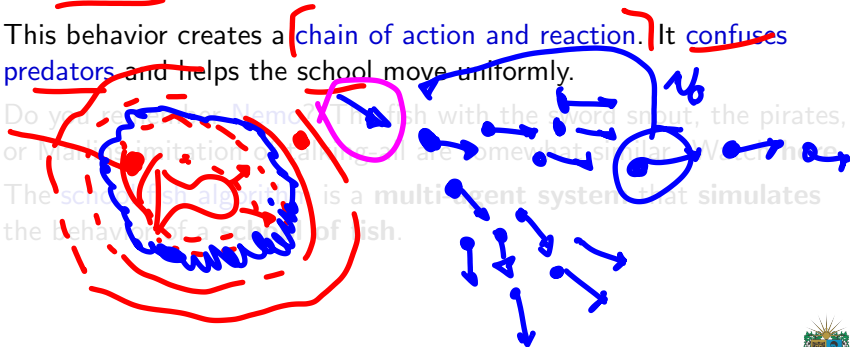
- **School fish** are interesting. When a predator attacks, they become confused by the large number of individuals and their **diverse movements**.

- The idea is simple: *"Don't touch me, don't come too close, but stay somewhat close."*
- This behavior creates a chain of action and reaction. It confuses predators and helps the school move uniformly.
- Do you remember Nemo the fish with the sword and the pirates, or Marlins imitation of a shark? It's somewhat similar. Watch [here](#).
- The school fish algorithm is a multi-agent system that simulates the behavior of a school of fish.



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FO's



Ant Colony Algorithm

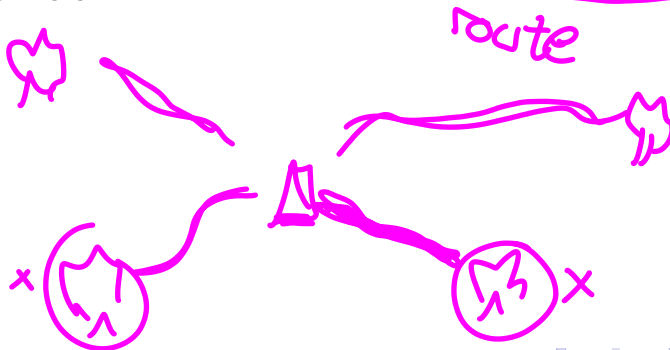
ACO

- The **ant colony algorithm** is a **multi-agent system** that **simulates** the behavior of an **ant colony**.
- The **ant colony algorithm** is based on the **social behavior** of **ants** and the use of **pheromones**. Watch here.
- The ant colony algorithm is used to solve optimization problems. Watch here.



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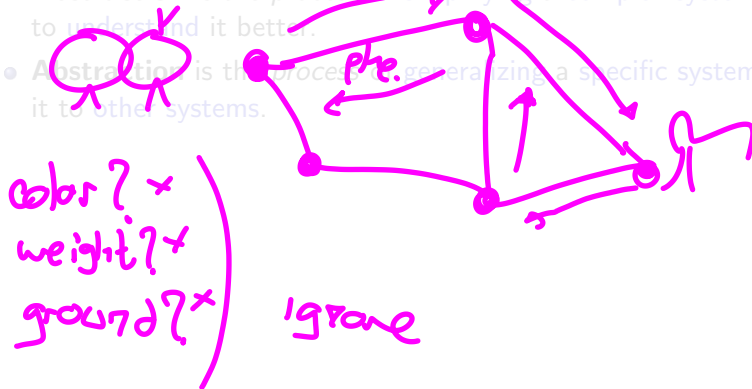
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What is Abstraction?

- **Abstraction** is the process of excluding minor details in order to focus on the important aspects of a system.

- Abstraction is the process of simplifying a complex system in order to understand it better.
- Abstraction is the process of generalizing a specific system to apply it to other systems.



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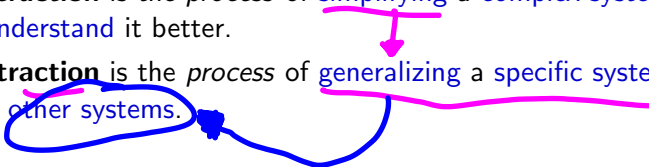
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O.O.P.



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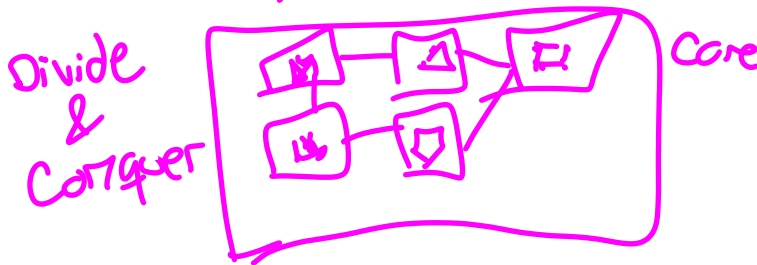
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- **Modularity** is the process of dividing a system into smaller parts called modules.

- **Modularity** is the process of organizing a system into independent units that can be developed and maintained separately.

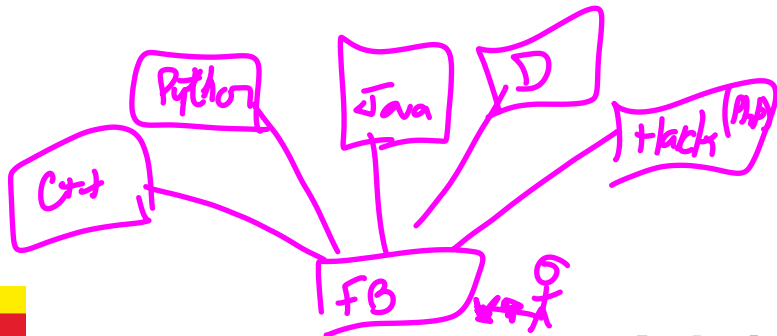
- **Modularity** is the process of reducing the complexity of a system by breaking it into smaller parts.

→ sub-systems
S.A.P.



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Abstraction and Modularity

- **Abstraction** and **modularity** are *two important concepts* in **systems analysis**.
- They help **reduce** the **complexity** of a **system** by **ignoring details** and **dividing** it into **smaller parts**.
- They help **improve** the **understanding**, **development**, and **maintenance** of a **system**.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

